ORBCOMM®

WHERE DATA DRIVES DECISIONS



CT 1010

Hardware Guide

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PREFACE

Purpose

This guide is as an overview of the hardware characteristics and specifications for the CT 1010.

Notation

Hardware components and hardware labels in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Bulleted lists highlight information where order or sequence is not crucial.

Device and Battery Safety Warnings

CAUTION: DO NOT short circuit or expose the battery to temperatures above the maximum rated temperature.

CAUTION: DO NOT throw the internal battery or the device into fire.

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: Store in a cool, well-ventilated area. Elevated temperatures will result in shortened battery life.

CAUTION: If shipping the device, contact your local shipping carrier for safe shipping guidelines.

Environmental Protection

Waste electrical products should not be disposed of with household waste. Please recycle where facilities exist. Check with our Local Authority or retailer for recycling advice.

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1 PRODUCT OVERVIEW

The CT 1010 (model number CT1010) container monitoring device combines telematics with sensor technology to protect cargo, reduce the risk of fires, identify mishandling, make containers safer for crews to operate, and deliver container visibility around the clock.

The CT 1010 features a container interior sensor pod that detects ambient temperature, light, motion, shock, rollover, door openings and closings, and more.

The CT 1010 is powered through an internal battery that is recharged using an integrated solar panel.

Figure 1: CT 1010



The CT 1010 includes the following features and functionality:

- Remotely programmable EPC Gen 2 compliant RFID tag for terminal gate and port crane integration
- Advanced power management for long battery life
- Wireless cellular and BLE
- 3-axis digital accelerometer
- Storage of more than 2000 messages (90+ days of operation)
- GPS, GLONASS, BeiDou, Galileo, and QZSS support
- Built-in cellular antenna, GPS/GNSS antenna, and BLE antenna
- Tamper detection sensor
- Ambient temperature sensor to measure internal and external temperatures for abnormal rates of change
- and much more

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2 SPECIFICATIONS

2.1 Temperature

Parameter	Value
Operating temperature range	-30°C to +70°C (-4°F to +158°F) excludes battery charging
Recommended storage temperature range	-20°C to +60°C (-4°F to +140°F)

2.2 Internal Battery Temperature

Parameter	Value
Battery charging temperature range	-30°C to +50°C (-4°F to + 122°F)
Battery discharge temperature range	20°C to +70°C (68°F to + 158°F)
Battery storage temperature range[1] (nominal)	-40°C to +85°C (-40°F to + 185°F)

2.3 Electrical

2.3.1 Power

The CT 1010 is powered through the internal battery that is recharged using an integrated solar panel.

2.3.2 Solar Panel (Battery Charging) Electrical Specifications

The solar panel provides power to the battery charging system.

Parameter	Min.	Max.	Units
Solar Open Circuit Voltage	4.84	-	V
Solar Power	550	-	mW
Charge Current	1	125	mA

2.3.3 Battery Voltage Measurement

The device can measure the input voltage over the range 2.5 V to 4.2 V.

2.3.3.1 Battery Electrical Specifications

The device includes a rechargeable long-life, industrial grade 3.6 V nominal battery. The battery pack is within the device enclosure.

[1]Storage at extreme temperatures can reduce battery life.

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Following are the electrical specifications for the battery.

Parameter	Min.	Typical	Max.	Units
Capacity	1000	1100	-	mAh
Voltage (nominal)	-	3.6	-	V
Maximum Discharge Current	2.0	-	-	A
Maximum Charge Current	200	-	-	mA

2.3.3.2 Battery Controller

Charge and discharge are controlled so that battery discharge is not allowed over the battery's safe discharge operating conditions.

Parameter	Typical	Max.	Units
Charge Current	-	180	mA
Float Voltage	4.05	-	V
Charge Time	-	8	Hours
Over Discharge Voltage	2.5	-	V
Over Charge Voltage	4.2	-	V
Over Discharge Current	2.5	-	A
Short Circuit Current	6	-	A

2.3.4 Current Consumption

Current values are at nominal battery voltage of 3.6 V.

Parameter	Min.	Units
Hibernation	40	µA
Idle[1]	100	µA
Cellular Transmit (pulsed)	2000	mA
Cellular Receive	45	mA
Application Processing	10	mA

2.4 RF Specifications

2.5 Cellular Communication

The device includes a cellular module cable of global cellular communication. Cellular communication is over LTE, 3G, and 2G networks.

Parameter	Value
LTE category	Cat 1

[1]This is average background current including sensor polling.

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Parameter	Value
LTE bands	1, 2, 3, 4, 5, 7, 8, 12, 13, 18, 19, 20, 25, 26, 28, 34, 38, 39, 40, 41
UMTS/HSPA+ Bands	1, 2, 4, 5, 6, 8, 19
GSM Bands	850/900/1800/1900 MHz

2.5.1 BLE RF Interface

The device includes an internal BLE antenna.

The BLE system allows the device to communicate as a peripheral with a mobile phone or other BLE enabled host. It also allows the device to communicate as a host to local wireless sensors.

The following table shows the BLE characteristics.

Parameter	Min.	Typical	Max.	Units
Frequency	2405	-	2480	MHz
Bluetooth version compliance	-	5.0	-	-
Receive sensitivity	-96	-	-	dBm
TX power	-	6	-	dBm

2.5.2 GNSS

A fully integrated GNSS solution that supports GPS, GLONASS, BeiDou, and Galileo.

The GNSS system includes an internal antenna. The manufacturer's specifications are shown in the following table.

Parameter	Typical
Time to First Fix	
Cold Start	30 s
Assisted	4 s
Hot Start	1 s
Sensitivity	
Tracking	-162 dBm
Hot Start	-159 dBm
Cold Start	-148 dBm
Accuracy	
Horizontal Position (CEP)	2 m

2.6 Sensors

The device includes the following sensors:

- Time of Flight Door Sensor - detects door position (open or closed). It is located in the sensor pod.
- Ambient Temperature Sensor - measures temperature over a range of -40°C to +85°C (-40°F to +185°F).
- IR Temperature Sensor - reports the temperature within the asset.

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- Ambient Light Sensor - detects visible light inside the asset. It is located in the sensor pod and detects light from 0.01 lux to 64k lux. The main use for this sensor is to detect and track potential damage to the asset (container), such as holes in the roof.
- Tamper Detection Sensor - detects the mounted state of the device. The sensor can detect when the device is mounted to ferrous metal, such as a steel shipping container.

2.7 Accelerometer

The device has two accelerometers to allow for dedicated functions that may require different settings. The first provides motion detections, and the second is for advanced functions such as shock and rollover detection.

The following table shows the specifications for both accelerometers.

Parameter	Condition	Min.	Typical	Max.	Units
Acceleration Range	software selectable	-	±2	-	g
		-	±4	-	g
		-	±8	-	g
		-	±16	-	g
Bandwidth Filtering	Selectable via digital interface	40	-	400	Hz
Sensitivity	2 g	-	256	-	LSB/g
	4 g	-	128	-	LSB/g
	8 g	-	64	-	LSB/g
	16 g	-	32	-	LSB/g

2.8 RFID Tag

The device includes an EPC Gen 2 compliant RFID tag (frequency: 860 to 960 MHz and 4 kbits EEPROM) to communicate with passive RFID readers.

2.9 Bluetooth Low Energy (BLE)

The BLE system allows the device to communicate as a peripheral with a mobile phone or other BLE enabled host. It also allows the device to communicate as a host to local wireless sensors.

The following table shows the BLE characteristics.

Parameter	Min.	Typical	Max.	Units
Frequency	2405	-	2480	MHz
Bluetooth version compliance	-	5.0	-	-
Receive sensitivity	-96	-	-	dBm
TX power	-	6	-	dBm

2.10 Non-Volatile Flash Storage

Parameter	Value
Non-Volatile Flash	8 MB, 100,000 write-erase cycles over its operating life.

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2.11 Internal Antennas

2.11.1 Cellular Antenna

Parameter	Value
Operating temperature range	-40°C to +85°C (-40°F to +185°F)
Frequency range	Low: 698 to 960 MHz, High: 1710 to 2690 MHz
Impedance	50 Ohm
VSWR	≤5.8
Gain	1.0 dBi max. 698 to 960 MHz, 2.3 dBi max.1710 to 2690 MHz

2.11.2 RFID Antenna

Parameter	Value
Operating temperature range	-40°C to +85°C (-40°F to +185°F)
Frequency range	860 to 960 MHz
Impedance	50 Ohm
VSWR	≤4.5:1
Gain (Phi realized gain)	-1.4 DBi max.
Polarization	Linear

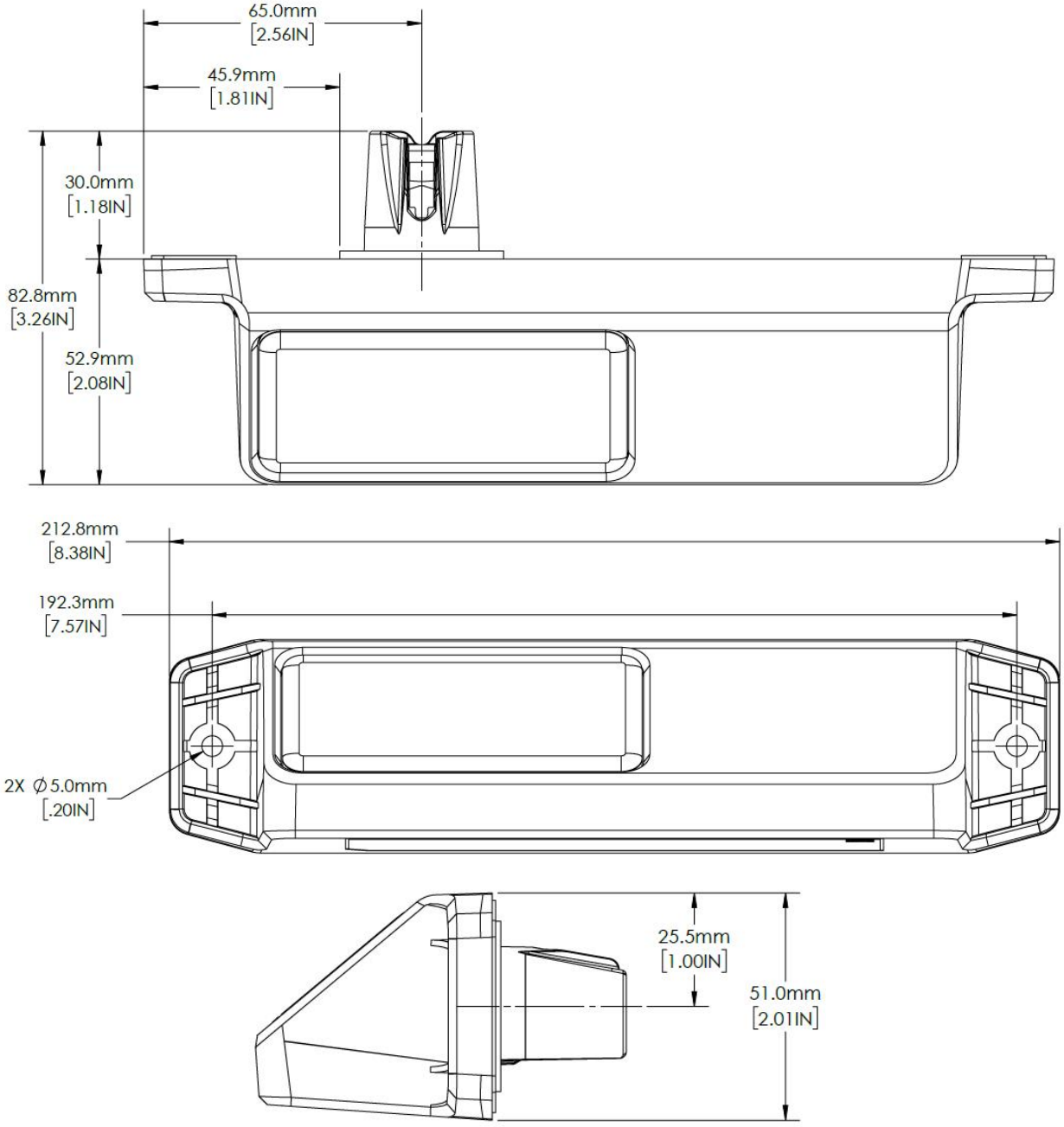
2.11.3 BLE Antenna

Parameter	Value
Operating temperature range	-40°C to +85°C (-40°F to +185°F)
Frequency range	2.4 to 2.485 GHz
Impedance	50 Ohm
VSWR	≤1.25
Gain (Phi realized gain)	1.36 DBi max.
Polarization	Linear

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2.12 Mechanical Dimensions

The device's mechanical dimensions are shown below.



Labeling

The device enclosure provides a location for the placement of a manufacturing label. Refer to APPENDIX 1 for ATEX labeling requirements.

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3 COMPLIANCE

ISED (Canada)

- IC: 11881A-BTCT1
- CONTAINS IC: 10224A-2022EG21GL
- ICES-003
- IC compliance statement

  This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) This device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

  Le present appareil est conforme aux CNR d'Industrie Canada applicable aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, meme si le brouillage est susceptible d'en compromettre le fonctionnement.

FCC (USA)

- FCC ID: XGS-BTCT1
- CONTAINS FCC ID: XMR202212EG21GL
- CFR 47 Part 25
- CFR 47 Part 15
- FCC compliance statement

  This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation. “

CE MARK (Europe)

- RED 2014/53/EU
- EU Declaration of Conformity

  Hereby, ORBCOMM Inc. declares that the radio equipment types listed in this document comply with Directive 2014/53/EU.

  The full text of the EU declaration of conformity is available from http://www2.orbcomm.com/eudoc.

RoHS

- Restriction of Hazardous Substances (RoHS) [1]

[1] European Union's (EU) Directive 2002/95/EEC "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.

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Anatel

AT&T

PTCRB

RF Exposure and Non-Modification Warning

RF Exposure statement This equipment complies with the radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20 cm (8 in.) between the radiator and any part of the human body.

Non-modification Warning statement Equipment changes or modifications not expressly approved by the party responsible for FCC compliance, could void the user's authority to operate the equipment and could create a hazardous condition.

Ingress Protection

- CT 1010 enclosure: IP69K
- CT 1010 enclosure and sensor pod: IP67

Country Approval

The device meets the importation plan to correspond to Evergreen depot /service locations. Some limitations are due to regulations in some countries.

- China
- Malaysia
- Singapore
- Taiwan
- Thailand
- Vietnam

Environmental

Parameter	Description
Vibration	The device meets all its specifications during exposure to random vehicular vibration levels per AAR-S-9401, section 3.2.4.2
Humidity	The device meets all its specifications during exposure to 90% relative humidity at +70°C (158°F), per the test methodology of SAE J1455, section 4.2.3 (3 x 8-hour humidity cycle per figure 4a)
Mechanical Shock	The device meetsMIL-STD-810H, section 516.8, Procedure 1, 40 g.
Thermal Shock	The device meets all its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2
Handling Drop	The device meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1.
High Pressure Wash	The device enclosure meets all its specifications after a pressure wash test as specified in ISO 20653 Table 7, IPx9k.

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Parameter	Description
Ingress - Dust and Sand	The enclosure provides protection against ingress of dust and sand as specified in IEC-60529, section 13.4 and 13.6.2 (IP6x).
Ingress Water	The device meets ingress protection to IPx7 as per IEC-60529, section 14.2.7.
Salt Spray Atmosphere	The device meets all its specifications after a salt spray test as detailed in SAE J1455, section 4.3.3.1.
Immersion	The device meets all its specifications after a 6-hour alternating hot/cold salt-water immersion test as detailed in SAE J1455, section 4.3.3.2.
Exposure to Chemicals and Oils	The device meets all its specifications after a light to moderate splash test as detailed in SAE J1455 section 4.4.3.2, for the following chemicals: Window Washer Solvent — Soap and Detergents Antifreeze Water Mixture — Axle Grease Brake Fluid — Diesel Exhaust Fluid Dust Control Agents (magnesium chloride) — Diesel Fuel Moisture Control Agents (calcium chloride) — Engine Oils and Additives Engine Degreaser — Ether Gasoline — Rear Axle Oil Fuel Additives — Alcohol Battery Acid — Waxes Kerosene — Freon Spray Paint — Paint Stripper Muriatic Acid — Ammonia
Fungus	The device meets all its specifications after a fungus test as detailed in SAE J1455, section 4.6.3.
Flammability	The device meets flammability rating: UL 94 V-0
ESD	The device meets all its specifications after exposure of the enclosure to 6 kV ESD contact discharge per IEC 61000--4-2, level 3.

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